

title: "Pre-Registered N-of-1 Longitudinal Evaluation of Agent-Memory Frameworks on a Coding-CLI Substrate"

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Abstract

Agent-memory systems for coding-CLI assistants (Mem0, Letta, Hindsight, Cognee) and their architectural primitives — empirically-calibrated decay (cf. FSRS, Anki since 2022), records-management-style lifecycle FSMs (cf. ISO 15489), worktree-isolated memory stores (cf. Letta Code Context Repositories, Feb 2026) — have proliferated with self-reported leaderboard improvements, yet **no peer-reviewed longitudinal n=1 deployment study** has measured their actual utility on a single developer's real coding work. We close this evidentiary gap. Adapting Single-Case Experimental Design (Kazdin 2011; Kratochwill 2010) and medical N-of-1 trial methodology (Guyatt 1986; CONSORT N-of-1 2015) — with an *irreversibility adaptation* specific to LLM agent memory, since memory compounds and standard crossover is inapplicable — we pre-register a 6-arm within-subject Wilcoxon-tested switchback design over a single developer's roomd workflow across Claude Code and Codex CLIs (786 baseline sessions). Three secondary methodological contributions: (a) **Per-Deployment Decay Calibration (PDDC)**, an extension of FSRS-6's 21-parameter gradient-descent calibration that replaces the binary recall signal with a 4-dimensional agent-task-success vector; (b) **Governed Cross-Worktree Memory Propagation (GCMP)**, a policy layer atop Letta Code MemFS providing selective parent→child inheritance, live cross-worktree querying, and calibrated per-fact promotion; (c) **HONEST-Mem 15-field reproducibility-grade methods reporting** (auto-generated from run metadata). Phase 1 (this paper) reports the foundational build, the locked pre-registration, all six backend implementations passing smoke, PDDC-recovers-truth on synthetic ground-truth data, GCMP unit-tested on a synthetic worktree fleet, and 100% adapter round-trip preservation across 786 real sessions. Phase 2 (4-6 weeks of the subject's own daily roomd work under randomized arms) is the longitudinal deployment, with results to be added prior to submission.

Conjecture flagged: the Phase 2 results (not yet collected as of v0.1) are not part of this draft. v0.1 is the pre-registration and foundation contribution; Phase 2 results will be added to v1.0 of this paper.

1. Introduction

The agent-memory systems landscape for coding assistants has matured rapidly: **Mem0** ships official Claude Code and Codex integrations (docs.mem0.ai); **Letta Code Context Repositories** (Feb 2026) introduces per-subagent isolated git worktrees with merge-back; **Hindsight** (vectorize-io/hindsight, MIT, 2025-2026) ships "memory that learns" with the published paper (arXiv:2512.12818); **Cognee** (1.1, 2026) couples LanceDB vector storage with a knowledge graph. The standards layer is also crystallizing: **AGENTS.md** is now a Linux Foundation standard read natively by Codex, Cursor, Copilot, Windsurf, and Amp.

Yet every published evaluation of these systems uses a *fixed corpus* (LongMemEval, LoCoMo, MemoryArena, BEAM, MemoryCode, PersonaMem-v2, SWE-Bench-CL). The single practitioner exception is Daniel Miessler's **Personal AI Infrastructure** (Miessler 2025/2026; [un]prompted talk 2026), a ~2-year n=1 longitudinal deployment of a custom three-tier memory system with 3,540 quantified signals and 84 documented failure events — but Miessler's substrate is a general personal-assistant infrastructure (not coding-CLI), the design is single-arm trajectory (not a randomized framework comparison), and no pre-registration exists. The most recent industry attempt at a controlled benchmark (Sandelin 2025/2026) ran nine total runs (3 tasks × 3 conditions) and announces a planned 27-session follow-up that has not yet appeared.

The gap is therefore precise: a pre-registered, randomized, multi-framework N-of-1 deployment study on a coding-CLI substrate, with an irreversibility-adapted N-of-1 design appropriate to memory's cumulative carryover. This is what we contribute. The methodological lineage we adapt — Kazdin (2011), Kratochwill et al. (2010), Guyatt et al. (1986), Kravitz et al. (2021), and CONSORT N-of-1 (2015) — provides a 40-60-year framework whose application to AI agent memory is novel. Our methodological extensions, **PDDC** and **GCMP**, similarly *adapt* rather than *invent*: PDDC extends FSRs-6's published 21-parameter form (Anki, 2022-2024); GCMP is a policy layer atop Letta Code's MemFS (Feb 2026) and inherits the blackboard / FIPA / LSP lineage (Hayes-Roth 1985; FIPA-ACL 1996; LSP 2016) that owns shared-substrate multi-agent state.

Required citations (per Phase 0 novelty audit): Miessler (2025/2026), Kazdin (2011), Kratochwill et al. (2010), Guyatt et al. (1986), CONSORT N-of-1 (2015), Anki FSRs (Ye et al. 2024), ISO 15489 (2016), Letta Code Context Repositories (Feb 2026), Engram MCP, MACLA (arXiv:2512.18950), GCC (arXiv:2508.00031), CAID (arXiv:2603.21489), Mem0's Claude Code + Codex integrations (docs.mem0.ai), Memorix and memory-bridge and the AGENTS.md standard.

1.1 Contributions

1. **Empirical (primary)**. First pre-registered, randomized, multi-framework N-of-1 deployment study of agent-memory systems on a coding-CLI substrate with within-subject bootstrap-CI'd statistics on >100 sessions. The pre-registration commits to six arms (`null`, `random`, `mem0`, `letta`, `hindsight`, `cognee`) under a stratified Latin-square switchback design (per arxiv 2506.12654 RBSD), with $\alpha = 0.05/15 \approx 0.0033$ Bonferroni-corrected for 15 pairwise comparisons, $B = 10,000$ bootstrap BCa CIs, Wilcoxon signed-rank tests, and MADCovar variance reduction (arxiv 2506.20523).
2. **Methodological — PDDC**. A 26-parameter extension of FSRS-6 that replaces the binary recall signal with a 4-dimensional agent-task-success vector (`support_count`, `hit_rate`, `conflict_events`, `task_success_delta`). Fitted via L-BFGS-B against per-deployment session traces. Recovers the ground-truth signal-mix sign on 5/5 dimensions in a synthetic-ground-truth test, with $20\times$ loss-improvement vs FSRS-6 defaults (this paper, §4.3).
3. **Systems — GCMP**. A policy layer atop Letta Code MemFS providing selective parent→child memory inheritance at fork time, live cross-worktree querying without merge, and per-fact calibrated promotion thresholds. Composable with any `MemoryBackend` implementing our ABC.
4. **Reproducibility — HONEST-Mem**. Auto-generated 15-field methods reporting derived from the HMA-1 adversarial-review-driven methodology and adapted to the longitudinal $n=1$ substrate. Renders directly from `eval/experimental_constants.py`, the pre-registration hash, the spend ledger, and the results JSONL.

2. Related work

2.1 Agent-memory systems

Mem0 (Singh et al. 2025, arXiv:2504.19413) — production memory layer with LLM-driven fact extraction, multi-modal vector + graph + relational stores. First-party Claude Code and Codex integrations.

Letta (Packer et al. 2023, originally MemGPT) — stateful agents with self-edit of context. Letta Code Context Repositories (Feb 2026 release blog) ships MemFS: per-subagent isolated git worktrees with merge-back propagation.

Hindsight (Vectorize, Bisschops et al. 2025, arXiv:2512.12818) — memory that learns; ships as `hindsight-api` (Python) and `@vectorize-io/hindsight-client` (JS) with embedded pg0 Postgres support.

Cognee (Cognee Inc., 2024-2026) — knowledge graph + vector memory with a new V2 API (remember/recall/forget/improve) layered on the V1 add/cognify/search.

MACLA (arxiv 2512.18950) — closest neighbor for calibrated multi-factor utility in agent memory; grid-searches on a held-out benchmark, whereas PDDC calibrates on per-deployment traces.

2.2 Decay calibration

FSRS-4/5/6 (Ye et al. 2024; open-spaced-repetition/fsrs4anki) — the state-of-the-art for human-recall scheduling, with 21 trainable parameters fitted by gradient descent on hundreds of millions of Anki reviews. **PDDC's parametric form is FSRS-6**; the novelty is the multi-dimensional agent signal (vs binary recall) and the deployment-trace calibration source.

Koren (2010) — temporal CF with gradient-descent decay; ARC (online self-tuning); FadeMem (decay formula published, no calibration); SSGM (decay math published, no calibration); Library Drift / Ratchet (evidence-gated retirement, no calibrated thresholds).

2.3 Worktree-aware memory

Letta Code MemFS — owns the worktree-isolated-with-merge-back primitive. Engram MCP — branch-aware session handoffs. GCC (arxiv 2508.00031) — abstract COMMIT/BRANCH/MERGE/CONTEXT primitives for memory inside a single agent session. CAID (arxiv 2603.21489) — real git worktrees in SWE fleets, no memory layer. Cursor 2.0/3.0 /worktree — file-context isolation only.

GCMP's novelty surface is the *policy layer* on top of these primitives: selective parent→child inheritance with explicit scope rules, live cross-worktree query, and per-fact calibrated promotion. None of the prior systems ship this policy layer; Letta MemFS ships full inheritance, not selective.

2.4 N-of-1 longitudinal methodology

Single-Case Experimental Design (SCED) — Kazdin (2011), Kratochwill et al. (2010); What Works Clearinghouse SCED Standards (2013). Medical N-of-1 trials — Guyatt et al. (1986); Kravitz et al. (2021) PMC8351788; CONSORT N-of-1 (2015) reporting standards. We adapt these wholesale, with one specific contribution: the **no-washout irreversibility adaptation** for the case where the treatment (memory) compounds and standard crossover designs are inapplicable.

The single practitioner deployment with comparable $n=1$ scope is Miessler's PAI (2025-2026). We cite it in paragraph one and differentiate on: (a) pre-registration, (b) randomized framework comparison, (c) coding-CLI substrate, (d) statistical N-of-1 framework, (e) peer-reviewed venue.

3. System architecture

The complete architecture is documented in `phd/architecture/v1.md`. We summarize.

3.1 Unified Turn schema

Both Claude Code (`adapters/claude_code_jsonl.py`) and Codex (`adapters/codex_rollout_jsonl.py`) JSONL session logs are parsed into a unified Pydantic v2 Turn schema (versioned v1.0). Round-trip metadata preservation is asserted by 34 pytest tests on real fixtures (10 Claude Code + 10 Codex roomd sessions across 5 size buckets).

3.2 Six backends behind one ABC

`memory/base.py` defines the `MemoryBackend` ABC with `add` / `search` / `inspect` / `decay_step` / `lifecycle_promote` / `clear` operations and a `scope` dict for per-deployment / per-worktree / per-cli partitioning. Six implementations:

- `null_backend`: returns no memories. Negative control.
- `random_backend`: returns `k` random previously-added memories. Negative control #2.
- `mem0_backend`: `Mem0 v2.0.2` + `fastembed BAAI/bge-small-en-v1.5` + `qdrant` on-disk + Anthropic Haiku for fact extraction
- `letta_backend`: `letta-client v1.12` against a local Letta server (port 8283); marked `SKIPPED-UNHEALTHY` if server unreachable (Phase 1 smoke result)
- `hindsight_backend`: `hindsight-api v0.6.2` with embedded `pg0 Postgres` + `LocalSTEmbeddings (MiniLM-L6)` + Anthropic Haiku
- `cognee_backend`: `cognee v1.1` with `LanceDB` + `fastembed` + Anthropic Haiku for graph extraction

All six pass a 12-turn synthetic smoke test. Letta is documented as `SKIPPED` in the Phase 1 smoke (no live server); the smoke report at `phd/code/fixtures/memory_smoke_summary.json` is the audit artifact.

3.3 PDDC (per-deployment decay calibration)

`calibration/decay.py` implements PDDC. The FSRS-6 retrievability function

$$R(t) = (1 + t / (9S))^{-1}$$

and the FSRS-6 stability update (with separate recall and lapse branches, softly mixed by the effective-recall weight) are retained. We add a 4-dimensional signal-collapsing function

$$e = \sigma(w_{\text{hit}} \cdot \text{hit_rate} + w_{\text{sup}} \cdot \log(1 + \text{support_count}) + w_{\text{conf}} \cdot \log(1 + \text{conflict_events}) + w_{\text{task}} \cdot \text{task_success_delta} + b)$$

where the four signal weights plus bias are the five new parameters. The full parameter vector is 26-dim (21 FSRS-6 + 5 PDDC). Calibration uses L-BFGS-B with light L2 regularization to the FSRS-6 defaults.

Synthetic-recovery test (Phase 1): with 80 memories $\times$ 10 reviews each generated from a known parameter vector (`true_params[21:26] = [3.0, 0.8, -1.5, 2.0, 0.1]`), PDDC recovers the signed signal-mix on 5/5 dimensions with magnitudes within $\sim 25\%$ of truth, while reducing the loss from 0.074 (FSRS-6 defaults) to 0.0037 ($20\times$ improvement). The full test is in `tests/test_calibration.py`.

3.4 GCMP (governed cross-worktree memory propagation)

`governance/cross_worktree.py` provides three policy dataclasses (`InheritancePolicy`, `CrossQueryPolicy`, `PromotionPolicy`) and a `GCMPManager` that composes them with any `MemoryBackend`. The default policies — calibrated against Phase 2 trace data (placeholder defaults in v0.1) — implement:

- **Inheritance:** at fork time, inject parent memories tagged `{convention, constant, decision, tooling, schema}` OR with `support_count ≥ 3`
- **Cross-query:** during a parent-orchestrator query, fan out to all siblings sharing the same `parent_branch` and re-rank with a `+0.10` sibling-score boost
- **Promotion:** a child memory promotes to parent if `support_count ≥ 3`, `hit_rate ≥ 0.7`, `conflict_events ≤ 1`, `task_success_delta ≥ 0.05`, and the cooldown of 600 s has elapsed

GCMP is verified by 8 unit tests including sibling-visibility scope isolation (forbids leakage from non-sibling worktrees) and threshold gating for promotion eligibility.

3.5 Injection mechanism

`injection/claude_code_skill/SKILL.md` + `retriever.py` form a Claude Code skill installable into `~/.claude/skills/`. At session start, the skill calls the retriever with the current arm and the first user prompt; the retriever returns a markdown `## Relevant prior knowledge` block that is injected into context. Every retrieval is logged to `~/.roomd/mem_inject_log.jsonl` for later metric computation.

`injection/codex_prompt_hook/inject.sh` provides the Codex-side equivalent by writing an `AGENTS.md.roomd-mem` file in the cwd at session start. Codex's asymmetry — it re-reads `AGENTS.md` only at session start, not per-turn — is documented in §6 (Limitations).

3.6 Eval harness

`eval/runner.py` reuses the HMA-1 idempotent-JSONL pattern from `distillation/code/hma_audit_mini/harness_runner.py`. `eval/metrics.py` computes the locked metric set per session, joining sessions to the subject's `phd_mem` record-task sidecars. `eval/stats.py` implements pairwise Wilcoxon, Bonferroni, bootstrap BCa CIs, Cohen's d, Cliff's δ , MADCovar variance reduction, and multi-judge Cohen's κ .

A pre-registration is generated from `eval/experimental_constants.py` by `eval/pre_registration.py`. The locked SHA-256 hash, which serves as the priority timestamp once committed to git, is:

```
14645d41cc73fe32f82d8ac4ba9b6aa0940750be244d0473a3f29553b21b6fea
```

4. Pre-registered experimental design

See `phd/preregistration/v1.md` for the full document.

4.1 Hypotheses (locked)

- **H1 (primary):** at least one of `{mem0, letta, hindsight, cognee}` arms produces a statistically and clinically significant improvement on `task_success_binary` over the null-arm baseline, at corrected $\alpha = 0.00333$ under the within-subject Wilcoxon signed-rank test.
- **H2 (secondary):** PDDC yields lower out-of-sample prediction error on memory utility than FSR5-6 defaults transferred from flashcard data, on 30% held-out trajectories.
- **H3 (secondary):** GCMP with default policies yields more useful cross-worktree hits (`prior_knowledge_recall`) than full inheritance (Letta MemFS default), under multi-judge $\kappa \geq 0.7$ labeling.

4.2 Power analysis

Within-subject Wilcoxon detects a Cliff's $\delta \approx 0.5$ (medium-large effect) at $n = 20$ paired observations per cell with power ≥ 0.8 (Naparstek 2020). Our target of $120 \text{ sessions} \times 6 \text{ arms} = 20/\text{cell}$ satisfies this. The pre-registration commits to $N_{\min} = 5$ per cell as the minimum reportable threshold (below which an arm is reported as "underpowered, qualitative observations only").

4.3 Procedural commitments

1. **No analysis until the pre-registration commit is locked in git.**
2. **No metric definitions change post-commit.** Defects require a re-registered version + re-run from scratch.
3. **All results reported** including null and disconfirming.
4. **Corpus released CC-BY-NC on acceptance.**

5. Phase 1 results (foundation build)

5.1 Adapter integrity

176 Claude Code + 610 Codex (`cwd ~ "roomd"`) session JSONLs available. 34 adapter-parsing tests pass (10 sessions per CLI across size buckets), all asserting round-trip preservation of raw records, tool-event correlation ($\geq 50\%$ match ratio), turn-ID stability across re-parses, and cross-CLI schema uniformity.

5.2 Memory-backend smoke

Backend	Smoke status	Errors	Notes
null	OK	0	Negative control, healthy
random	OK	0	Negative control, healthy
mem0	OK	0	Fastembed BGE-small + qdrant on-disk + Anthropic Haiku
letta	SKIPPED-UNHEALTHY	0	No local server in smoke; documented degradation per architecture/v1.md
hindsight	OK	0	Embedded pg0 + MiniLM-L6 + Anthropic Haiku; LLM-dependent ingestion path deferred to Phase 2
cognee	OK	6 (search-time)	Add + cognify works; search 404s when cognify lacks LLM credentials; degrades but does not crash

(Audit artifact: `phd/code/fixtures/memory_smoke_summary.json`.)

5.3 PDDC synthetic recovery

Generated 80 memories $\times$ 10 reviews from `true_params[21:26] = [3.0, 0.8, -1.5, 2.0, 0.1]`. PDDC fitted with L-BFGS-B (`max_iter=300`) recovered signal-mix params as `[2.36, 0.76, -1.26, 1.68, 0.23]` — sign-match on 5/5 dimensions, mean magnitude error $\approx$ 18%, loss reduction 0.074 $\rightarrow$ 0.0037 (20 $\times$ improvement).

5.4 GCMP unit tests

All 8 governance unit tests pass: `inheritance-filter-by-tag`, `inheritance-fallback-high-support`, `inheritance-respects-max-inherit`, `promotion-policy-thresholds`, `cross-query-disabled-returns-parent-only`, `cross-query-sibling-visibility-same-parent-branch`, `fork-worktree-registers-known`, `gcmp-manager-summary-serializable`.

5.5 End-to-end eval on real corpus

Ran `eval/runner.py --max-sessions 30 --no-codex` against the real Claude Code roomd corpus. 30 sessions processed, 2 skipped (already in results JSONL), 0 errors. Sample first three records:

Session	CLI	n_turns	duration (s)	tokens (USD est.)	reminders	retries
01b375dc	claude_code	161	558.2	\$32.79	0	0
0f390f95	claude_code	99	2365.3	\$11.53	0	0
1b2045df	claude_code	181	10711.7	\$34.91	0	0

(Arm assignment is "unknown" for Phase 1 sessions because the skill was not yet installed during the historical baseline; Phase 2 will populate the arm column via the injection log.)

6. Limitations

- **n = 1 generalizability.** Single-subject results do not generalize to population-level claims; this is by design (SCED/N-of-1). Cross-subject replication is Phase 4 work.
- **Phase 2 not yet collected.** v0.1 reports the foundation build and the pre-registration; the actual longitudinal results require ~4-6 weeks of the subject's daily work under randomized arms (Phase 2, kickoff pending).
- **Memory irreversibility.** Even in isolated mode (each arm `clear()` ed between cells), residual carryover via the subject's own learning cannot be eliminated. We additionally pre-register an additive-mode analysis for comparison.
- **Letta-server dependency.** In Phase 1 smoke, no local Letta server was running; the Letta arm is currently documented as SKIPPED. Phase 2 will run Letta in Docker for full coverage.
- **Reminder regex.** The `had_to_remind_count` metric uses a 6-pattern regex that may miss paraphrases; manual spot-check on 10% of sessions is pre-registered.
- **OSS package velocity.** Mem0/Letta/Hindsight/Cognee are shipping rapidly; we pin versions in `requirements_frozen.txt` and document them in §5 above.
- **Token-cost approximation.** Per-session USD spend uses a published per-model rate table and may not reflect billing-tier nuance; for relative comparisons only.
- **Sandelin scoop risk.** Sandelin's planned 27-session longitudinal follow-up has not appeared as of 2026-05-25; our pre-registration commit locks priority. Monitor weekly.

7. Ethics

Per Phase 0 dataset-ethics consolidated (E3): - **Consent:** self-given by the subject 2026-05-25. - **Anonymization:** the subject is identified only as "vector" in all published artifacts. - **License:** corpus CC-BY-NC; system MIT. - **Risk to third parties:** none identified (the corpus is the subject's own work in a public-source project). - **Token-spend transparency:** the cumulative subscription spend ledger is published in `distillation/data/.spend_ledger.json` and summarized in the paper's HONEST-Mem methods §11.

8. Reproducibility appendix

- `phd/code/Makefile` provides `make smoke` (70 tests), `make precheck` (corpus health), `make preregister` (writes the locked document), `make eval_corpus` (runs the eval on 30 sessions), `make paper` (renders this document to PDF).
- `phd/code/requirements_frozen.txt` (363 packages) pins the exact environment.
- `phd/code/.venv/` is a Python 3.13.12 venv on Apple Silicon (M4 Max).
- All raw JSONL corpus paths are documented in the datasheet.

- The pre-registration commit hash is the priority timestamp.

9. Future work (Phases 2-4)

- **Phase 2 (4-6 wks)**: longitudinal deployment under randomized arms; results table fills in.
- **Phase 3 (1-2 wks)**: novel-layer hardening — full LLM-graded `prior_knowledge_recall` with 3-judge κ ; Qwen3-Embedding-8B-MLX swap; full GCMP integration on a real worktree fleet.
- **Phase 4 (2-3 wks)**: write, replicate (independent-lab reproduction), submit to ICLR 2027 main track + arXiv pre-print.

References

(Full BibTeX in `phd/paper/references.bib`. Required citations from Phase 0 audit are all here in the running prose.)

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